

Name: ANSWERS

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SCH3U: Unit Test: Quantities in Chemical Reactions

Formulas:

$$n = \frac{m}{M}$$

$$n = \frac{N}{N_A}$$

$$\text{Conversion factor} = \left(\frac{\text{want}}{\text{have}} \right)$$

$$\% \text{ mass} = \frac{m_{\text{part}}}{m_{\text{whole}}} \times 100\%$$

$$\% \text{ yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

$$N_A = 6.022 \times 10^{23}$$

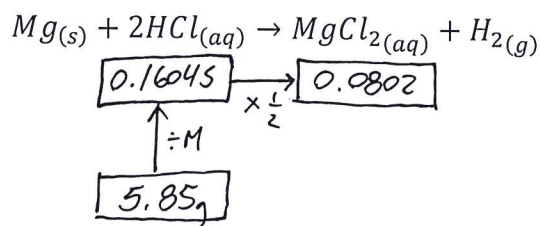
$$\text{Molecular Multiplier} = \left(\frac{m_{\text{molecular}}}{m_{\text{empirical}}} \right)$$

Multiple Choice Answers:

1 B	2 D	3 A	4 C	5 D
6 B	7 D	8 A	9 C	10 C
11 B	12 D	13 B	14 D	15 C
16 A	17 B	18 B/A	19 D	20 D

Part B: Application Questions

1. If 5.85 g of HCl are consumed in the following reaction, how many moles of MgCl_2 are expected? [3]



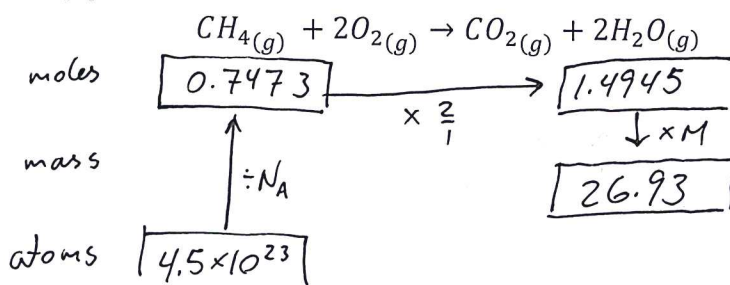
$$M_{\text{HCl}} = 36.46 \text{ g/mol}$$

$$\begin{aligned} n_{\text{HCl}} &= \frac{m}{M} \\ &= \frac{5.85\text{g}}{36.46 \text{ g/mol}} \\ &= 0.16045 \text{ mol} \end{aligned}$$

$$\begin{aligned} n_{\text{MgCl}_2} &= (0.16045 \text{ mol}) \left(\frac{1}{2} \right) \\ &= \boxed{0.0802 \text{ mol}} \end{aligned}$$

∴ 0.0802 mol are expected

2. If 4.50×10^{23} molecules of CH_4 are burned, in sufficient oxygen, how many grams of H_2O are expected? [4]



$$M_{\text{H}_2\text{O}} = 18.02 \text{ g/mol}$$

$$\begin{aligned} n_{\text{CH}_4} &= \frac{N}{N_A} \\ &= \frac{4.50 \times 10^{23} \text{ molecules}}{6.022 \times 10^{23} \frac{\text{molecules}}{\text{mol}}} \\ &= 0.7473 \text{ mol} \end{aligned}$$

$$\begin{aligned} n_{\text{H}_2\text{O}} &= (0.7473 \text{ mol}) \left(\frac{2}{1} \right) \\ &= 1.4945 \text{ mol} \end{aligned}$$

$$\begin{aligned} m_{\text{H}_2\text{O}} &= nM \\ &= (1.4945 \text{ mol}) (18.02 \text{ g/mol}) \\ &= \boxed{26.9 \text{ g}} \end{aligned}$$

∴ Approximately 26.9 g of H_2O are expected

3. Determine the percent composition of CuSO_4 . [3]

$$\begin{aligned} \% \text{Cu} &= \frac{63.55 \text{ g/mol}}{159.61 \text{ g/mol}} \times 100\% \\ &= 39.8\% \end{aligned}$$

$$\begin{aligned} \% \text{S} &= \frac{32.07 \text{ g/mol}}{159.61 \text{ g/mol}} \times 100\% \\ &= 20.1\% \end{aligned}$$

$$\begin{aligned} \% \text{O} &= \frac{4(18.02 \text{ g/mol})}{159.61 \text{ g/mol}} \times 100\% \\ &= 40.0\% \end{aligned}$$

$$M_{\text{CuSO}_4} = 159.61 \text{ g/mol}$$

$$M_{\text{Cu}} = 63.55 \text{ g/mol}$$

$$M_{\text{S}} = 32.07 \text{ g/mol}$$

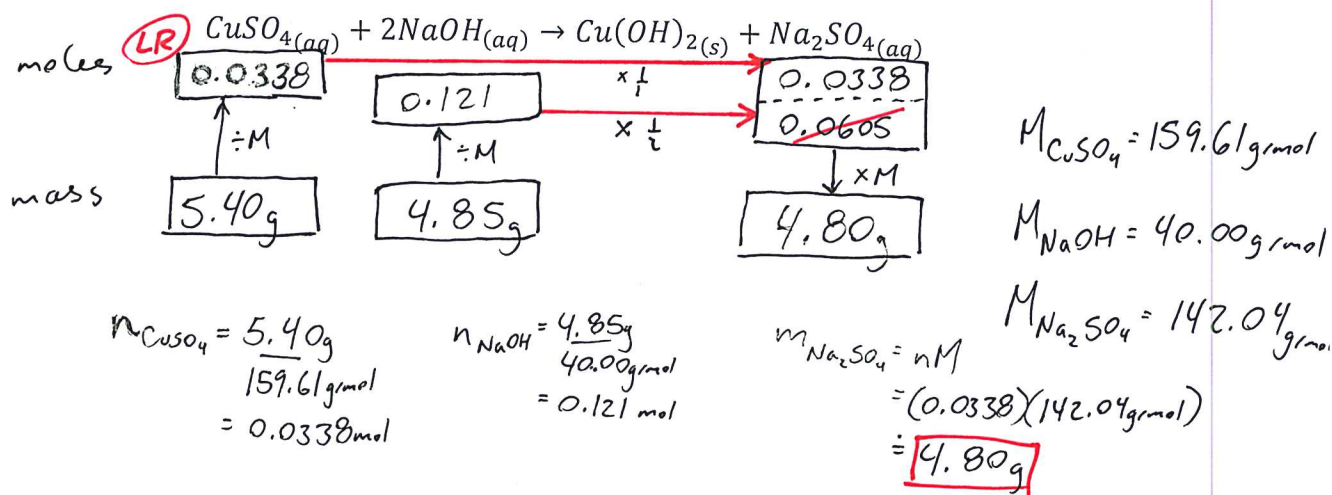
$$M_{\text{O}} = 16.00 \text{ g/mol}$$

∴ The mass percent composition of CuSO_4 is

$$\begin{array}{l} \boxed{39.8\% \text{ Cu}} \\ \boxed{20.1\% \text{ S}} \\ \boxed{40.0\% \text{ O}} \end{array}$$

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4. If 5.40 g of CuSO_4 are reacted with 4.85 g of NaOH  $\therefore 4.80\text{g}$ of Na_2SO_4 are expected

a. Which reactant will be limiting? [1]

 CuSO_4 b. How many grams of Na_2SO_4 are expected? [3] 4.80g c. If the reaction has a 90.0% yield, how many grams of Na_2SO_4 are produced? [2]

$$\text{actual} = (0.90)(\text{theoretical})$$

$$= (0.90)(4.80\text{g})$$

$$= 4.32\text{g} \leftarrow \text{will accept 3 sig digs as well}$$

$$\doteq \boxed{4\text{g}}$$

$\therefore$ The reaction yields approx 4.32g of Na_2SO_4

5. A compound contains C, H and O. The compound has a percent composition of 40.00 % C, 6.71 % H and 53.29 % O.

a. Determine the empirical formula [3]

Assume a 100g sample;

$$n_{\text{C}} = \frac{40.00\text{g}}{12.01\text{g/mol}} = 3.33\text{mol}$$

$$n_{\text{H}} = \frac{6.71\text{g}}{1.01\text{g/mol}} = 6.64\text{mol}$$

$$n_{\text{O}} = \frac{53.29\text{g}}{16.00\text{g/mol}} = 3.33\text{mol}$$

C	H	O
3.33	6.64	3.33
$\div 3.33$	$\div 3.33$	$\div 3.33$
1	2	1

 $\therefore$ The empirical formula is $\boxed{\text{CH}_2\text{O}}$

b. If the compound has a molar mass of 150.15 g/mol, what is the molecular formula? [2]

$$\text{Molecular multiplier} = \frac{M_{\text{compound}}}{M_{\text{empirical}}}$$

$$= \frac{150.15\text{g/mol}}{30.03\text{g/mol}} = 5$$

$$M_{\text{CH}_2\text{O}} = 30.03\text{g/mol}$$

 $\therefore$ The molecular formula is $\boxed{\text{C}_5\text{H}_{10}\text{O}_5}$

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6. A 1.50 g sample of hydrocarbon, containing C and H, undergoes complete combustion to produce 4.40 g of CO_2 and 2.70 g of H_2O . Analysis with a mass spectrometer indicates a molar mass of approximately 30 g/mol. What is the molecular formula of the compound? [6]

Empirical Formula;

$$n_{\text{C}} = \frac{m_{\text{CO}_2}}{M_{\text{CO}_2}} \left(\frac{1 \text{ C}}{1 \text{ CO}_2} \right)$$

$$= \frac{4.40 \text{ g}}{44.01 \text{ g/mol}}$$

$$= 0.100 \text{ mol}$$

$$n_{\text{H}} = \frac{m_{\text{H}_2\text{O}}}{M_{\text{H}_2\text{O}}} \left(\frac{2 \text{ H}}{1 \text{ H}_2\text{O}} \right)$$

$$= \frac{2.70 \text{ g}}{18.02 \text{ g/mol}} \left(\frac{2}{1} \right)$$

$$= 0.2997 \text{ mol}$$

$$\approx 0.300$$

C	H
0.100	0.300
$\div 0.100$	$\div 0.100$
1	3

$\therefore$ The empirical formula is CH_3

Molecular Formula;

$$\text{Molecular multiplier} = \frac{M_{\text{compound}}}{M_{\text{empirical}}}$$

$$= \frac{30 \text{ g/mol}}{15.03 \text{ g/mol}}$$

$$\approx 2$$

$\therefore$ The molecular formula is $\boxed{\text{C}_2\text{H}_6}$

BONUS: In question 4, how many grams of the excess reactant will remain after the reaction is completed? (theoretical, not based on percent yield) [1]

$\boxed{2.15 \text{ g}}$

Amt of NaOH used;

$$n_{\text{NaOH}} = n_{\text{CuSO}_4} \left(\frac{2}{1} \right)$$

$$= (0.0338 \text{ mol}) \left(\frac{2}{1} \right)$$

$$= 0.0676 \text{ mol}$$

$$m_{\text{NaOH}} = (0.0676 \text{ mol}) (40.00 \text{ g/mol})$$

$$= 2.704 \text{ g}$$

$$\text{Amt of excess NaOH} = 4.85 \text{ g} - 2.704 \text{ g} = 2.15 \text{ g}$$